Assignment 2

CSE 251

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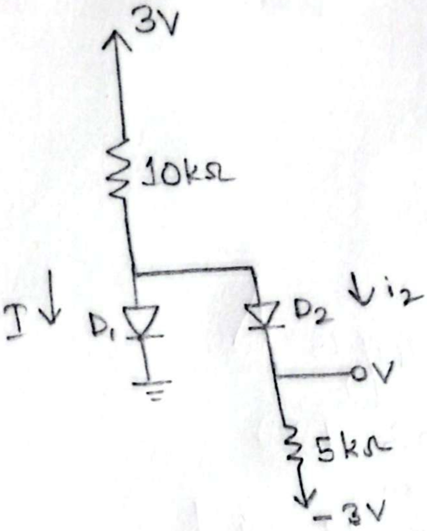
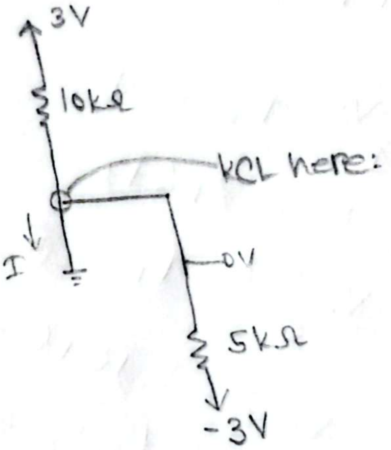
7b)



lets assume, D1 = on, D2 = on



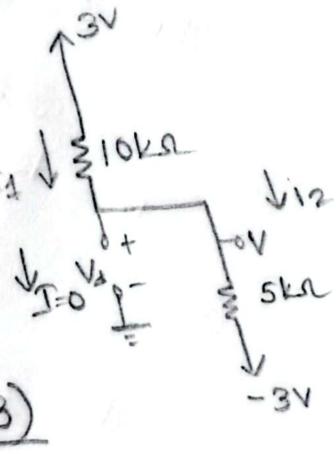
$V = 0V$

$$\frac{3-0}{10} = I + \frac{0-(-3)}{5}$$

$\frac{3}{10} - \frac{3}{5} = I = -\frac{3}{10}$, since current is negative, D_1 cannot be on,

$i_2 = \frac{0-(-3)}{5} = \frac{3}{5}$, D_2 may be on

lets assume D_1 is off and D_2 is on



$$\frac{3-V}{10} = \frac{V-(-3)}{5}$$

$$\frac{3-V}{10} = \frac{V+3}{5}$$

$$\frac{3-V}{2} = V+3$$

$$3-V = 2V+6$$

$$-3 = 3V$$

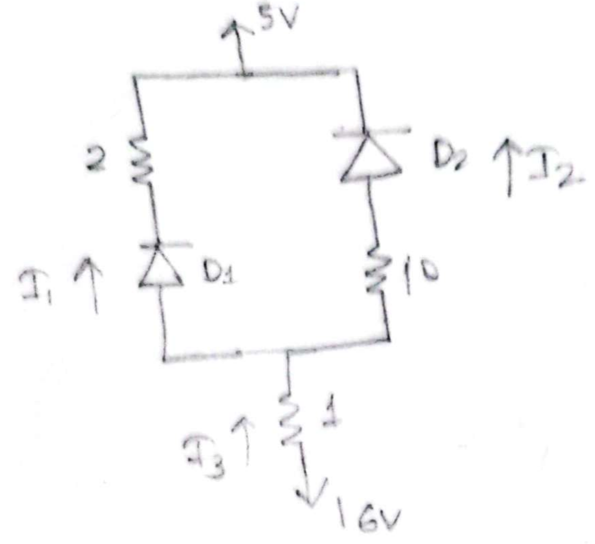
$$V = -1$$

$i_2 = \frac{-1-(-3)}{5} = \frac{2}{5}$, so D_2 may be on

$V_d = -1-0$
$= -1$, so D1 may be off.

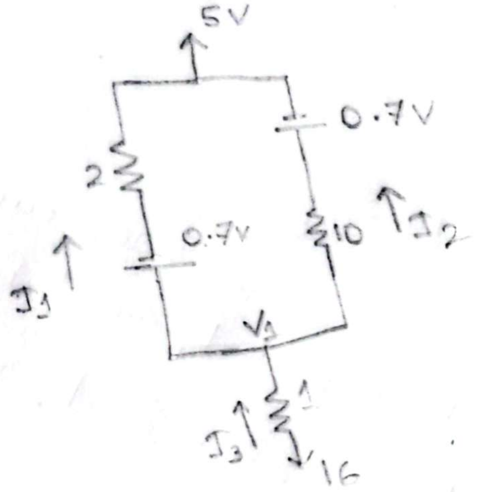
since both assumptions correct, D1 = off, D_2 = on,
$I = 0, V = -1V$

10



$V_{DO} = 0.7V$

lets assume, D_1 and D_2 on.



$I_3 = I_2 + I_1$

$$\frac{16 - V_1}{1} = I_3 \quad - (i)$$

$$V_1 - 5 = 0.7 + 2I_1$$

$$\frac{V_1 - 5.7}{2} = I_1 \quad - (ii)$$

$$V_1 - 5 = I_2 10 + 0.7$$

$$\frac{V_1 - 5.7}{10} = I_2 \quad - (iii)$$

$$16 - V_1 = \frac{V_1 - 5.7}{10} + \frac{V_1 - 5.7}{2}$$

$$16 - V_1 = \frac{V_1 - 5.7 + 5V_1 - 28.5}{10}$$

$$160 - 10V_1 = 6V_1 - 34.2$$

$$194.2 = 16V_1$$

$$V_1 = 12.1375$$

$$I_1 = \frac{12.1375 - 5.7}{2}$$

$$= 3.21875 \text{ mA}$$

$$I_2 = \frac{12.1375 - 5.7}{10}$$

$$= 0.64375 \text{mA}$$

$$I_3 = 3.8625 \text{mA}$$

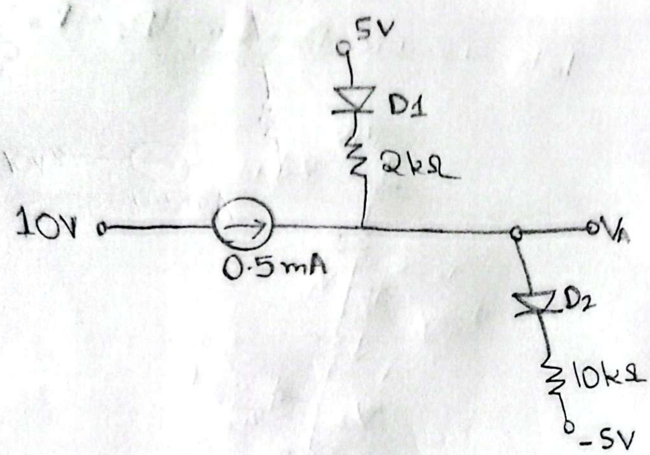
since both I_1 and $I_2 > 0$,
assumptions are correct.

$I_1 = 3.22 \text{mA}$

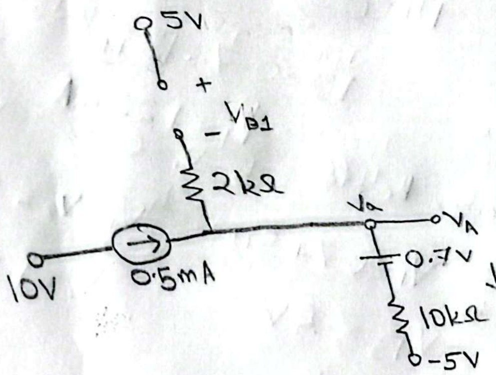
$I_2 = 0.64 \text{mA}$

$I_3 = 3.86 \text{mA}$

16.



lets assume, D_1 = off , D_2 = on



$V_{D_0} = 0.7V$

$V_a + 5 = 0.7 + I10$

$\frac{V_a + 4.3}{10} = I$

KCL at Va

$0.5 + 0 = \frac{V_a + 4.3}{10}$

$5 = V_a + 4.3$

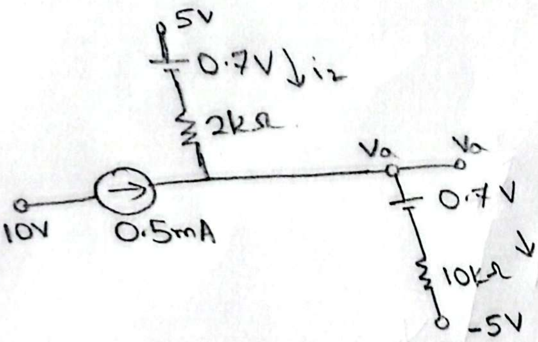
$V_a = 0.7$

$i_1 = \frac{0.7 + 4.3}{10} = 0.5$, so

D_2 may be on

$V_{D1} = 5 - 0.7$
$= 4.3V$, this is greater than 0.7V, so this assumption is wrong.

Assuming
D_1, D_2 = on



$0.5 + i_2 = i_1$

$5 - V_a = 0.7 + i_2 \times 2$

$\frac{4.3 - V_a}{2} = i_2$

$0.5 + \frac{4.3 - V_a}{2} = \frac{V_a + 4.3}{10}$

$0.5 = \frac{V_a + 4.3}{10} + \frac{V_a - 4.3}{2}$

$0.5 = \frac{V_a + 4.3 + 5V_a - 21.5}{10}$

$5 = 6V_a - 17.2$

$6V_a = 22.2$

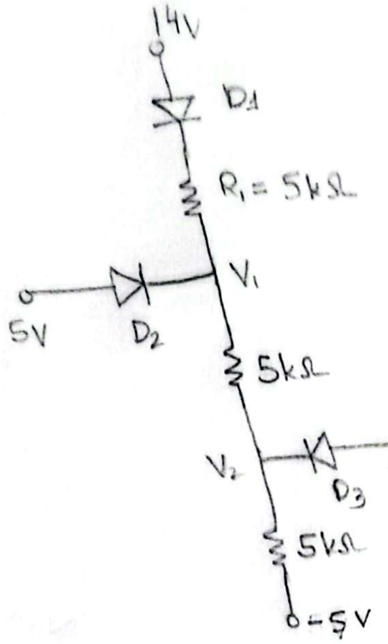
$V_a = 3.7V$

$i_2 = 0.3mA$

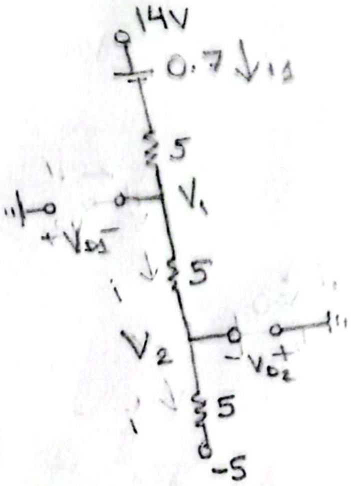
$i_1 = 0.8mA$

∴ both assumptions correct and $V_a = 3.7V$, D_1, D_2 = on

18.



lets assume, D_1 = on, $D_2 = D_3$ = off



$$14-(-5) = 0.7 + 5i_1 + 5i_1 + 5i_1$$

$$19 - 0.7 = 15 i_1$$

$$i_1 = 1.22\text{mA}$$

$i_1 > 0$, so D1 may be on,

$0 - V_1 = -7.2 < 0.7$, so D_2 may be off

$14 - V_1 = 0.7 + 5i_1$ $\quad$ $V_1 - V_2 = 5i_1$

$0 - V_2 = -1.1V < 0.7$, so D_3 maybe off.

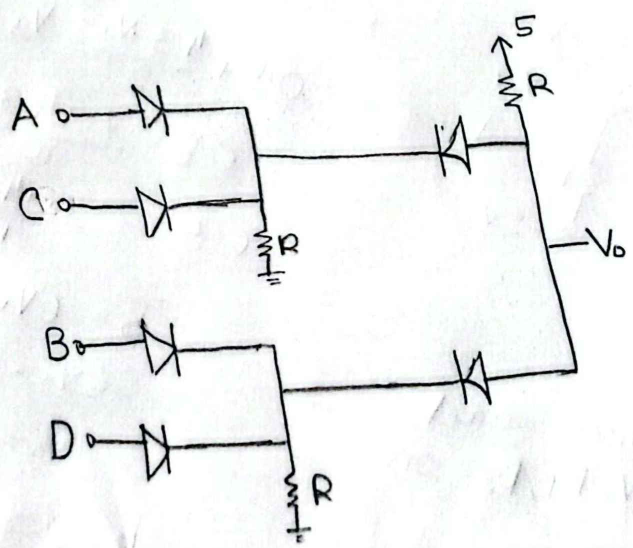
$V_1 = 7.2V$ $\quad$ $V_2 = 1.1$

Since all assumption correct

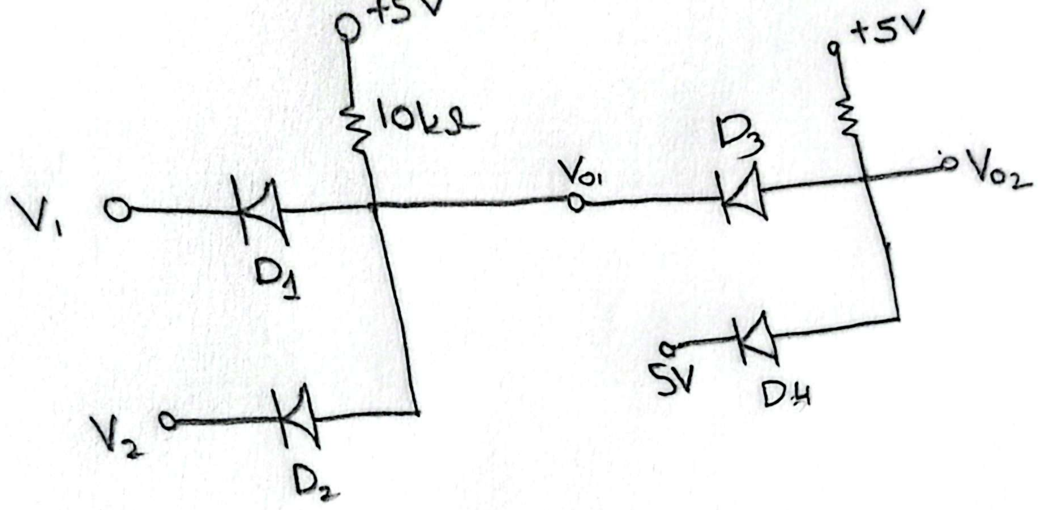
$V_1 - V_2 = 6.1V$

D_1 = on, $D_2 = D_3$ = off, $R_2 V = 6.1V$ (Ans)

1iii) $(A+C) \cdot (B+D)$



10.



$V_1 = 3V$ $\qquad V_{D_o} = 0.7V$

$V_2 = 1V$

$V_{o1} = \text{Min}(3+0.7, 1+0.7)$

$= 1.7$

$V_{o2} = \text{Min}(5+0.7, 1.7+0.7)$

$= 2.4V$

$V_{o2} = (V_1 \cdot V_2) \times 1$

$= V_1 \cdot V_2$